

Hermitian Matrix



Hermitian Matrix is a special matrix; etymologically, it was named after a French Mathematician Charles Hermite (1822 – 1901), who was trying to study the matrices that always have real Eigenvalues (<https://byjus.com/maths/eigen-values/>). The Hermitian matrix is pretty much comparable to a symmetric matrix. The symmetric matrix is equal to its transpose, whereas the Hermitian matrix is equal to its conjugate transpose, sometimes referred to as tranjugate.

The Hermitian matrix has complex numbers; however, its diagonal entries are real. The Eigenvalues of a Hermitian matrix are always real.

Let us learn more about Hermitian matrices and their properties in detail, along with hermitian matrix examples.

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What is a Hermitian Matrix?

A Hermitian matrix is a matrix that is equal to its conjugate transpose. Mathematically, a Hermitian matrix is defined as

A square matrix $A = [a_{ij}]_{n \times n}$ such that $A^* = A$, where A^* is the conjugate transpose of A ; that is, if for every $a_{ij} \in A$,

$$\overline{a_{ij}} = a_{ji}$$

($1 \leq i, j \leq n$), then A is called a Hermitian Matrix.

Matrix A is Hermitian Matrix $\Leftrightarrow A^* = A$

For example, let

$$A = \begin{bmatrix} 3 & 1-i \\ 1+i & -2 \end{bmatrix}$$

Then conjugate of A =

$$\overline{A} = \begin{bmatrix} 3 & 1+i \\ 1-i & -2 \end{bmatrix}$$

and transpose of conjugate of A =

$$A^* = \begin{bmatrix} 3 & 1-i \\ 1+i & -2 \end{bmatrix} = A$$

Therefore, $A^* = A$

Hence, A is a Hermitian Matrix.

- Note:** i. Every diagonal element of a Hermitian Matrix is a real number.
ii. Every real symmetric matrix is Hermitian.

Properties of a Hermitian Matrix

Let us take a note of some important properties of a Hermitian matrix.

- A square matrix A of order n is Hermitian if and only if every $a_{ij} \in \mathbb{R}$,
 $\overline{a_{ij}} = a_{ij}$
 $(1 \leq i, j \leq n)$.
- If A is a Hermitian matrix, and k is any real scalar, then kA is also a Hermitian matrix.

Let us take $(kA)^* = kA^* = kA$ {since k is a real number}

- If A is a Hermitian matrix, the $(A^*)^* = A$

As $(A^*)^* = (A)^* = A^* = A$.

- If A and B are two additional conformable Hermitian matrices, then $A + B$ is also Hermitian.

Given A and B are Hermitian, then

$$(A + B)^* = (\overline{A} + \overline{B})' = (\overline{A})' + (\overline{B})'$$

=

$$A^* + B^*$$

$\Rightarrow (A + B)^* = A + B$ is Hermitian.

- If A and B are square matrices, then $(AB)^* = B^*A^*$. If A and B are Hermitian, then $(AB)^* = BA$.

- The determinant of a Hermitian matrix is real.
- The inverse of a Hermitian matrix is Hermitian as well.
- Conjugate of a Hermitian matrix is also Hermitian.
- If A is Hermitian, then A^*A and AA^* is also Hermitian.
- Any square matrix can be represented as $A + iB$, where A and B are Hermitian matrices.
- For any square matrix A , if $A^* = -A$, then A is called the skew-Hermitian matrix. Any square matrix can be uniquely represented as a sum of a Hermitian and skew-Hermitian matrix.

Let A be any square matrix, then

$$A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*); \text{ where } (A + A^*) \text{ is Hermitian and } (A - A^*) \text{ is skew-Hermitian.}$$

- If A is Hermitian matrix, then A^n is also Hermitian for all positive integers n .

Given A is Hermitian, that is, $A^* = A$

$$\text{Now, } (A^n)^* = (A^*)^n = A^n$$

$$\Rightarrow (A^n)^* = A^n \text{ is Hermitian.}$$

- Trace of a Hermitian matrix is always real.

Eigenvalues of Hermitian Matrix

The Eigenvalues of a Hermitian matrix are always real.

Let A be a Hermitian matrix such that $A^* = A$ and λ be the eigenvalue of A .

Let X be the corresponding Eigen vector such that $AX = \lambda X$ where

$$X = \begin{bmatrix} a_1 + ib_1 \\ a_2 + ib_2 \\ \vdots \\ a_n + ib_n \end{bmatrix}$$

Then X^* will be a conjugate row vector. Multiplying X^* on both side of $AX = \lambda X$ we have,

$$X^*AX = X^*\lambda X = \lambda(X^*X) = \lambda(a_1^2 + b_1^2 + \dots + a_n^2 + b_n^2)$$

Clearly, $(a_1^2 + b_1^2 + \dots + a_n^2 + b_n^2)$ is real.

Now, $(X^*AX)^* = X^*A(X^*)^* = X^*AX$, hence X^*AX is Hermitian of order 1.

Thus X^*AX is a real number, consequently λ is also real.

Let us take an example,

$$A = \begin{bmatrix} 3 & 1-i \\ 1+i & 2 \end{bmatrix}$$

Clearly, $A^* = A$, A is Hermitian.

The characteristic polynomial of A is

$$|A - \lambda I| =$$

$$\begin{vmatrix} 3-\lambda & 1-i \\ 1+i & 2-\lambda \end{vmatrix}$$

$$= (3-\lambda)(2-\lambda) - [(1-i)(1+i)]$$

$$= \lambda^2 - 5\lambda + 4$$

$$= (\lambda - 1)(\lambda - 4)$$

Thus, the eigenvalues of A are 1 and 4 which are real.

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Solved Examples on Hermitian Matrix

Example 1:

Check whether the given matrix

$$A = \begin{bmatrix} 1 & i & 3i \\ -i & 0 & 2-i \\ -3i & 2+i & -1 \end{bmatrix}$$

is Hermitian.

Solution:

Given,

$$A = \begin{bmatrix} 1 & i & 3i \\ -i & 0 & 2-i \\ -3i & 2+i & -1 \end{bmatrix}$$

Conjugate of $A =$



$$\overline{A} = \begin{bmatrix} 1 & -i & -3i \\ i & 0 & 2+i \\ 3i & 2-i & -1 \end{bmatrix}$$

Conjugate transpose of A = $A^* =$

$$A = \begin{bmatrix} 1 & i & 3i \\ -i & 0 & 2-i \\ -3i & 2+i & -1 \end{bmatrix}$$

Hence $A^* = A$

Thus, A is Hermitian.

Example 2:

If k is complex number and A be a Hermitian matrix. Will kA be Hermitian?

Solution:

Given A is a Hermitian matrix, $A^* = A$ and k is any complex number

Now,

$$(kA)^* = \bar{k}A^* = \bar{k}A \neq kA$$

Since k is a complex number, and it is not equal to its conjugate.

Hence, $(kA)^* \neq kA$

Thus, kA is not a Hermitian matrix.

Frequently Asked Questions on Hermitian Matrix

Q1 How do you know if a matrix is Hermitian?

For the given matrix, find its conjugate and then transpose the conjugate matrix. If the transpose of the conjugate matrix is same as the given matrix, then the matrix is Hermitian.

Q2 How a Hermitian matrix different than a symmetric matrix?

If a matrix is equal to its transpose then it is called a symmetric matrix whereas if a matrix is equal to its conjugate transpose then it is a Hermitian matrix. Moreover, a Hermitian matrix have complex numbers whereas a symmetric matrix has real entries.

Q3 What type of eigenvalues do a Hermitian matrix have?

A Hermitian matrix always has real eigenvalues.

Q4 What is the determinant of a Hermitian matrix?

The determinant of a Hermitian matrix is real in nature.

Q5 Is addition of two Hermitian matrix again a Hermitian?

Yes, addition of two Hermitian matrix is again Hermitian.

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